# Ken (Tongli) Su

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CS PhD student working on **LLM agents, alignment, interpretability, and safety**, especially multi-agent collaboration and theory of mind: agents that reason about collaborators' beliefs, adapt to human values, and know when to seek human input. Co-first author of **Software World Models** (predicting what a change breaks before it runs) and **LARGER** (beats 7 baselines incl. Claude Code, Codex); first author, EMBC 2026 oral. Reviewer, NeurIPS 2026.

## Education

**Emory University** — Atlanta, GA
*Ph.D. in Computer Science* — Aug 2025 – Dec 2028 (expected)

**Columbia University** — New York, NY
*M.S. in Data Science, GPA: 3.95/4.00* — Dec 2023

**Emory University** — Atlanta, GA
*B.A. in Computer Science & B.A. in Economics (Dual Major), GPA: 3.84/4.00* — May 2022

## Selected Publications & Research

[1] **T. Su**[*], Y. Hu[*], L. Zhao, B. Zhu, J. Somasundaram, H. Haque. "Software World Models: From Consequence Prediction to Decision Value." *arXiv:2610.04940*, 2026. *(Under review; [*]equal contribution.)*

- Formulated **system-level consequence prediction** for coding agents: predict the **blast set** — the downstream services, libraries, and datastores a locally correct change will break — *before* execution, beyond what the agent itself observes.
- Built **Explore–Learn–Act**: execute counterfactual changes from restored system states, train Qwen2.5-Coder with **GRPO** on executed outcomes (plus an R-GCN graph-token variant with **SC-GRPO**), and act on per-consumer break probabilities.
- Blast-set F1 **0.431** → **0.571** vs. static reachability and ranking regret **more than halved**; on held-out real libraries, agent return **0.453** → **0.613**, above the CWM and Qwen-AgentWorld world-model baselines.

[2] Y. Hu, **T. Su**, Y. Sun, S. Ma, Q. Tan, L. Zhao. "Adaptive Memory Index Evolution for Agents." *arXiv:2610.34977*, 2026. *(Under review.)*

- **AMIE**: a plug-in **memory index evolution** layer that distills an LLM teacher's keep/suppress judgments into a Qwen3-8B LoRA policy and accumulates them into persistent per-memory access state, with no change to stored content or host retraining. Improved downstream F1/accuracy in **all 15** comparisons (**5 memory systems** incl. Mem0, A-MEM × **3 benchmarks**) by up to **25%** relative.

[3] Y. Hu[*], **T. Su**[*], L. Zhao, B. Zhu, H. Haque. "LARGER: Lexically Anchored Repository Graph Exploration and Retrieval." *arXiv:2605.16352*, 2026. *(Under review; [*]equal contribution.)*

- Anchored a coding agent's own lexical search hits into a repository graph and expanded confidence-filtered neighborhoods *inside* its search loop (imports, call chains, code–test links) with no added agent actions; built **AST-based graph construction** end to end.
- **+13.9 Acc@5** and **+12.2 Recall@5** over the strongest of **7 baselines** (BM25, Agentless, LocAgent, SWE-agent, OpenHands, Claude Code, Codex) across **4 benchmarks** spanning localization, test generation, and codebase QA.

[4] **T. Su**, A. Rafiei, M. van Assen, R. Sameni, G. D. Clifford, F. Marzbanrad, N. Katebi. "Cross-Modal Generative Framework for Signal Translation from Fetal–Maternal Electrocardiograms to Fetal Doppler Waveforms." *IEEE EMBC 2026*. **Accepted — Oral Presentation.**

- Designed a cross-modal generator (dilated convolutions + **cross-modal attention** + self-attention) for ECG-to-Doppler translation: PSD MSE **51% below** baseline; ablations show naive concatenation adds nothing while cross-modal attention cuts error **39%**.

[5] **T. Su**, N. Barari, N. Katebi. "A Review of Multimodal Pretraining Strategies for Physiological Wearable Sensor Foundation Models." *Under review*, 2026. — Taxonomy of five multimodal pretraining paradigms over **22 papers** across PPG/ECG/EEG/ACC.

## Industry Experience

**Research Intern** | Causal Dynamics Lab — May 2026 – Aug 2026

- Researched coding agents and code intelligence; built a **software world model** for multi-repository microservice systems that predicts which downstream services, libraries, and datastores an agent's change will break *before* it reaches CI/CD — failures local tests miss.
- Engineered a sandboxed **execution harness** that restores frozen system baselines, applies one patch, and reruns unit, service-health, and end-to-end checks across Go/Node/Python **gRPC microservices** and real library ecosystems (Marshmallow, OpenTelemetry).
- Built the training and serving pipeline: typed cross-repo dependency graphs (services, endpoints, schemas) fed to the LLM as graph tokens, **SFT + GRPO** on executed outcomes, and a **pre-merge risk check** that lets the agent rank changes, migrate consumers before shipping, and run costly integration checks only when worthwhile — ∼**40% fewer** services left broken on held-out systems.

**Machine Learning Engineer, GenAI** | DuPont Specialty Products USA, LLC — New York, NY | Jul 2024 – Aug 2025

- Built a lead identification pipeline on **Azure Databricks** (Spark, Delta Lake, generative models) behind an internal sales co-pilot that became the company's primary prospecting tool (**$60M+** estimated value); fine-tuned **Sentence-BERT** with triplet loss for retrieval and ranking: **Recall@5 0.86**, **MRR 0.64**, **+27%** over the incumbent GPT-based system with hallucinations eliminated.
- Engineered an async, load-balanced **OpenAI API** enrichment module: runtime 10 h → 6 h (**40%**), token costs **–$12K/yr**.
- Turned user feedback into model and feature improvements over a six-week sprint: **DAU +48%**, user satisfaction 4.0 → 4.6/5.

## Technical Skills

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| **Agents & LLMs** | LLM & coding agents, world models, agent memory, RL post-training (GRPO), fine-tuning (LoRA, distillation), retrieval/ranking (RAG, BM25, graph retrieval), Hugging Face Transformers, Sentence-BERT, OpenAI API, agent evaluation |
| **ML & Deep Learning** | PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost; graph neural networks (R-GCN), attention, cross-modal & generative models, self-supervised pretraining, time-series & biosignals |
| **Languages & Infra** | Python, SQL, R, Java, C++, Bash; Spark, PySpark, Pandas, Databricks, Airflow, PostgreSQL; AWS, Azure, GCP |